

JEE MAIN REVISION MASTER: WAVES 2026 1

QUESTIONS 1.67% WEIGHTAGE

Subject: Physics | Target Scope: JEE Main Previous Years | Empirical Base: 9 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
High	Kcet Waves - Core Application	9 PYQs	General Concept

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
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3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Kcet Waves - Core Application (High Yield — 9 PYQs)

Core Principle: Key problem pattern in Kcet Waves.

■ ■ Caution / JEE Pitfall: Watch for algebraic and sign errors.

■ Speed Trick: Verify dimensions and boundary conditions.

Representative Source PYQs:

- [2026 Shift] Q#1 (MCQ): "A source of frequency ν gives 6 beats/second when sounded with a source of frequency 200 Hz. The second Harmonic of frequency 2ν ..." [Ans: B]
- [2025 Shift] Q#2 (MCQ): "The angle between the particle velocity and wave velocity in a transverse wave is [except when the particle passes through the mean position..." [Ans: N/A]
- [2024 Shift] Q#3 (MCQ): "A motor-cyclist moving towards a huge cliff with a speed of 18kmh^{-1} , blows a horn of source frequency 325 Hz. If the speed of the sou..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2026 (Q#1)

A source of frequency ν gives 6 beats/second when sounded with a source of frequency 200 Hz. The second Harmonic of frequency 2ν of the source gives 8 beats/second when sounded with a source of frequency 420 Hz. The value of ν is

- (A) 205 Hz
- (B) 206 Hz **[Correct]**
- (C) 195 Hz
- (D) 210 Hz

Correct Answer: B

Solution Analysis: For beats, the number of beats per second is equal to the difference in the two frequencies. Given that the source of frequency ν gives 6 beats/second with 200Hz : $|\nu - 200| = 6$ So, $\nu = 200 + 6 = 206\text{Hz}$ or $\nu = 200 - 6 = 194\text{Hz}$ Now the second harmonic of the source has frequency 2ν . It gives 8 beats/second with 420Hz , so $|2\nu - 420| = 8$ This gives $2\nu = 420 + 8 = 428$ or $2\nu = 420 - 8 = 412$ So, $\nu = 214\text{Hz}$ or $\nu = 206\text{Hz}$ The common value from both conditions is $\nu = 206\text{Hz}$ So the correct option is Option B: 206 Hz .